

The topography of Iapetus' leading side

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Received 12 December 2006; revised 15 May 2007

Available online 18 July 2007

Abstract

We have used Cassini stereo images to study the topography of Iapetus' leading side. A terrain model derived at resolutions of 4–8 km reveals that Iapetus has substantial topography with heights in the range of –10 km to +13 km, much more than observed on the other middle-sized satellites of Saturn so far. Most of the topography is older than 4 Ga [Neukum, G., Wagner, R., Denk, T., Porco, C.C., 2005. *Lunar Planet. Sci. XXXVI*. Abstract 2034] which implies that Iapetus must have had a thick lithosphere early in its history to support this topography. Models of lithospheric deflection by topographic loads provide an estimate of the required elastic thickness in the range of 50–100 km. Iapetus' prominent equatorial ridge [Porco, C.C., and 34 colleagues, 2005. *Science* 307, 1237–1242] reaches widths of 70 km and heights of up to 13 km from their base within the modeled area. The morphology of the ridge suggests an endogenous origin rather than a formation by collisional accretion of a ring remnant [Ip, W.-H., 2006. *Geophys. Res. Lett.* 33, doi:10.1029/2005GL025386. L16203]. The transition from simple to complex central peak craters on Iapetus occurs at diameters of 11 ± 3 km. The central peaks have pronounced conical shapes with flanking slopes of typically 11° and heights that can rise above the surrounding plains. Crater depths seem to be systematically lower on Iapetus than on similarly sized Rhea, which if true, may be related to more pronounced crater-wall slumping (which widens the craters) on Iapetus than on Rhea. There are seven large impact basins with complex morphologies including central peak massifs and terraced walls, the largest one reaches 800 km in diameter and has rim topography of up to 10 km. Generally, no rings are observed with the basins consistent with a thick lithosphere but still thin enough to allow for viscous relaxation of the basin floors, which is inferred from crater depth-to-diameter measurements. In particular, a 400-km basin shows up-domed floor topography which is suggestive of viscous relaxation. A model of complex crater formation with a viscoplastic (Bingham) rheology [Melosh, H.J., 1989. *Impact Cratering*. Oxford Univ. Press, New York] of the impact-shocked icy material provides an estimate of the effective cohesion/viscosity at 0.04 ± 0.01 MPa/ 0.6 ± 0.2 GPa s. The local distribution of bright and dark material on the surface of Iapetus is largely controlled by topography and consistent with the dark material being a sublimation lag deposit originating from a bright icy substrate mixed with the dark components, but frost deposits are possible as well.

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Keywords: Iapetus; Saturn, satellites; Satellites, shapes; Satellites, surfaces; Cratering

1. Introduction

Little has been known about the topography of Iapetus, but topography is important in understanding the geology of

the surface and the thermal evolution of the lithosphere. The Voyager spacecraft imaged Iapetus only from distant positions (8.5 km/pxl at best) and with poor radiometric resolutions (Smith et al., 1981, 1982), which has hitherto prevented detailed topographic studies. Limited topographic information could be inferred from stereo image pairs compiled to anaglyph images (Croft, 1991) wherein individual scarps were identified and the morphology of craters was discerned, but no quantitative topo-

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data. Although we are primarily concerned with the topography of Iapetus, it is instructive to see what the global shape looks like from our approach. Fig. 2 (left) shows a terrain model when the ground points are referenced to a sphere (see Section 2). This reveals an extremely non-spherical shape of Iapetus consistent with the findings from global shape modeling, according to which Iapetus is best approximated by a $747.4 \pm 3.1 \times 712.4 \pm 2.0$ km oblate spheroid (Thomas et al., 2008).

But topography requires heights measured relative to an equipotential surface. At this place we adopt a geometrically defined reference surface given by the oblate spheroid axes above. This surface is not at equipotential level under current conditions of Iapetus' synchronous rotation but it likely was when Iapetus rotated at a period of ~ 16 h early in its history (Thomas et al., 2008). Most of Iapetus' surface features were inferred to be older than 4 Ga (Neukum et al., 2005) and are likely to have formed during the early fast rotating period [Ip (2006) has mentioned that damping of Iapetus' spin rotation is expected to occur in a long rather than short time scale]. Thus the chosen height reference is physically plausible. Minor deviations from the true equipotential surface will not change the relief of the features significantly. The terrain model with points referenced to the spheroid is shown in Fig. 2 (right). It is the basis for the subsequent analysis.

The terrain models of Iapetus' leading side were compiled from 12 individual stereo pairs with horizontal resolutions of 4–8 km and vertical accuracies of 300–800 m. The trailing-side DTM (Fig. 2, top) is a compilation from 2 stereo pairs with horizontal resolutions of 5–10 km and vertical accuracies of 200–400 m. Features smaller than or within these resolution limits are expected to be smoothed.

4. Topographic features

4.1. General topographic characteristics

4.1.1. Observations

Inspection of the DTM reveals that Iapetus' surface is dominated by impact-related topography. Craters are ubiquitous on Iapetus including large impact basins (Fig. 3). The largest impact basin (Fig. 2, 35° N, 272° E; Fig. 3, I) is about 800 km in diameter and has rim topography of up to 10 km extending over scales of some hundreds of km (Fig. 2, bottom profile). The rim-to-bottom relief is 14 km at maximum. Another topographic landform is the equatorial ridge, which is tens of kilometers wide with heights of up to 13 km (see crosses in Fig. 2) within the modeled area (limb profiles show relief up to 20 km elsewhere, Fig. 5, top). The stratigraphic relationships suggest that both the 800 km basin and the equatorial ridge belong to the oldest features with likely ages of 4.4–4.5 Ga (Neukum et al., 2005).

(Neukum hat nur das "basin" untersucht)

The lowest points within the modeled area have heights down to about -10 km and are found at the bottoms of basins VI and VII (Fig. 3).

4.1.2. Implications

Thomas et al. (2008) have computed residual limb topography on Iapetus and on the other icy Saturn satellites. This has shown that Iapetus is exceptional in its large surface relief. Our contiguous terrain model now confirms these findings. Moreover, Iapetus' topography can be compared with topography obtained for Tethys (Giese et al., 2006b) and with Voyager-based elevation models of large impact features on the middle-sized icy satellites (Moore et al., 2004a). All of these show lower height ranges than exhibited by Iapetus.

Obviously, Iapetus has retained much of its impact-related topography and the (old) ridge topography. This implies a thick lithosphere at the time of formation to support the associated loads. If it were the case that Iapetus is not homogeneous, density contrasts are expected to be of minor importance in supporting topography because Iapetus' low density (1.08 g/cm^3 , Thomas et al., 2008) implies only a small rock mass ratio. The thickness of the lithosphere can be estimated from models of lithospheric deflection by imposed surface loads (Brotchie and Silvester, 1969). Here, we solve for a lithosphere thick enough to prevent subsiding of the observed rim loads of the 800 km basin. The underlying assumption is that this topography has not suffered a significant height loss by lithospheric deflection. Measurements of a rim-height dependence on crater diameter would allow identification of rim topography subsidence but such measurements do not exist yet. An empirical relation for complex lunar craters (Pike, 1977) predicts a rim height of 3.4 km for an 800-km crater which is much lower than the observed rim heights and thus may support the assumption. As a model load for lithospheric deflection, we considered a segment of rim topography. To simplify the problem, we approximated the load represented by the rim profile by a disc-shaped load with the same force (Fig. 4). Deflection profiles were then computed on the basis of equations given in Appendix A. These reveal that lithospheric thicknesses of 100–125 km are required to prevent more than 1 km of subsidence of the load. For the deflections to be less than 2 km lithospheric thicknesses of 40–60 km are required. Larger deflections are unlikely to have occurred because this would have formed a pronounced depression on the spherical shell in the surroundings of the load which is not observed. From this we conclude that likely lithospheric thicknesses more than 4 Ga ago range from 50–100 km.

4.2. The equatorial ridge

4.2.1. Observations

A set of profiles across the ridge (Fig. 5) shows that the morphology changes along the equator. The top profiles r2–r5 display a trapezoidal shape up to 70 km wide with upper flanking slopes of typically 15° , and a pronounced plateau. However, the higher resolution anaglyph image (Fig. 3) reveals that the plateau actually has finer-scale topography consisting of an inner ridge aligned with two outer ridges of about the same height and narrow gaps in between. Profiles r3 and r4 indicate a distinct change in their north-facing flanking slopes from 15° to 8° and 4° , respectively. This is also seen in the anaglyph image (e.g., the dashed profile). Moreover, the anaglyph image reveals

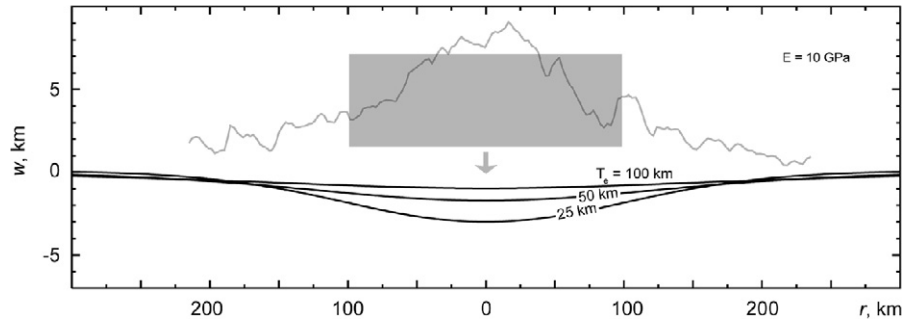


Fig. 4. Deflection w of the lithosphere as a function of radial distance r from the load (see Appendix A). For the purpose of modeling, the 8 km high load represented by the profile above (see Fig. 2 (right), dashed line) is replaced by a disc-shaped load exerting the same force. A lithosphere only 25 km thick is subject to a large deflection of 3 km but a lithosphere 100 km thick would be subject to a deflection smaller than 1 km for a Young's modulus (E) of 10 GPa. A weaker lithosphere with $E = 1$ GPa must be 125 km thick to prevent deflections larger than 1 km.

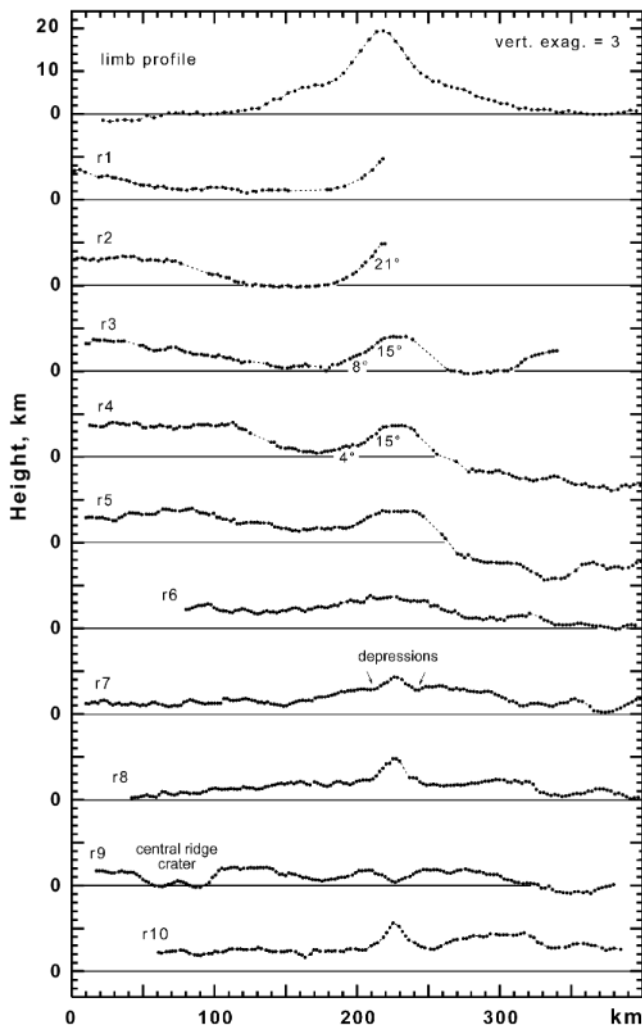


Fig. 5. Profiles across the equatorial ridge. Profile locations are shown in Fig. 6. The top profile is a section of limb profile N1482859953 (Thomas et al., 2008) showing ridge heights in a location distant from the area covered by the DTM.

that the pre-existing upper surface layer rose up along the flank of the ridge as it formed (arrows point to such places). There is a 100-km-wide depression with a sharp boundary north of the ridge and aligned with it (Fig. 2, longitudes $<250^\circ$ E). A similar southern depression may also exist as indicated by the DTM. Profile r7 marks a change in ridge shape. At this place the ridge

is made up of the inner ridge only, which exhibits a triangular shape there, but the outer ridges are at lower heights and less pronounced. The narrow gaps as observed above have become wider and form small depressions on both sides of the inner ridge (r7). Further away (r9) the ridge appears as a faint feature at the bottom of a 40-km-broad depression (see inset in Fig. 6), but along to the east (r10) it again appears as a triangular high-standing feature.

While the largest ridge heights reach 13 km within the DTM (Fig. 2) shape modeling has provided heights of up to 20 km in other places along the equator (Fig. 5, limb profile).

The trailing side DTM (Fig. 2, top) reveals that the ridge continues on this hemisphere.

4.2.2. Implications

The formation of the ridge is still an unresolved problem. Currently, there are two different theories. Porco et al. (2005) suggested an origin related to the overall shape or changes in spin state of Iapetus but they also noted that the ridge does not show characteristics or patterns comparable to the tectonic pattern seen in despinning models. In addition to this, Ip (2006) pointed out problems related to the old age of the ridge on the one hand and predicted long despinning time scales of Iapetus on the other hand. To avoid that problem among others, he suggested an exogenic origin where the ridge was formed by collisional accretion of a ring remnant. This theory invokes a ring remnant stemming from the formation of the proto-Iapetus, and a thick atmosphere to force the ring particles drifting towards the surface.

Our observations here argue for an endogenous origin of the ridge. The ridge's appearance is consistent with it having been formed by material rising from below, which thereby upwarped the surface on both sides. This is primarily suggested by the different slopes measured at the ridge flanks (r3, r4). The lower slopes are much less than the angle of repose and thus are potentially related to upwarping of the surface but the upper, steeper slopes are related to the rising material within this formational model. The observed broad depressions aligned with the ridge for longitudes $<250^\circ$ E (Fig. 2) may consequently be of flexural origin. If the ridge were to consist of piles of ring material slopes close to the angle of repose would be expected but this is not observed.

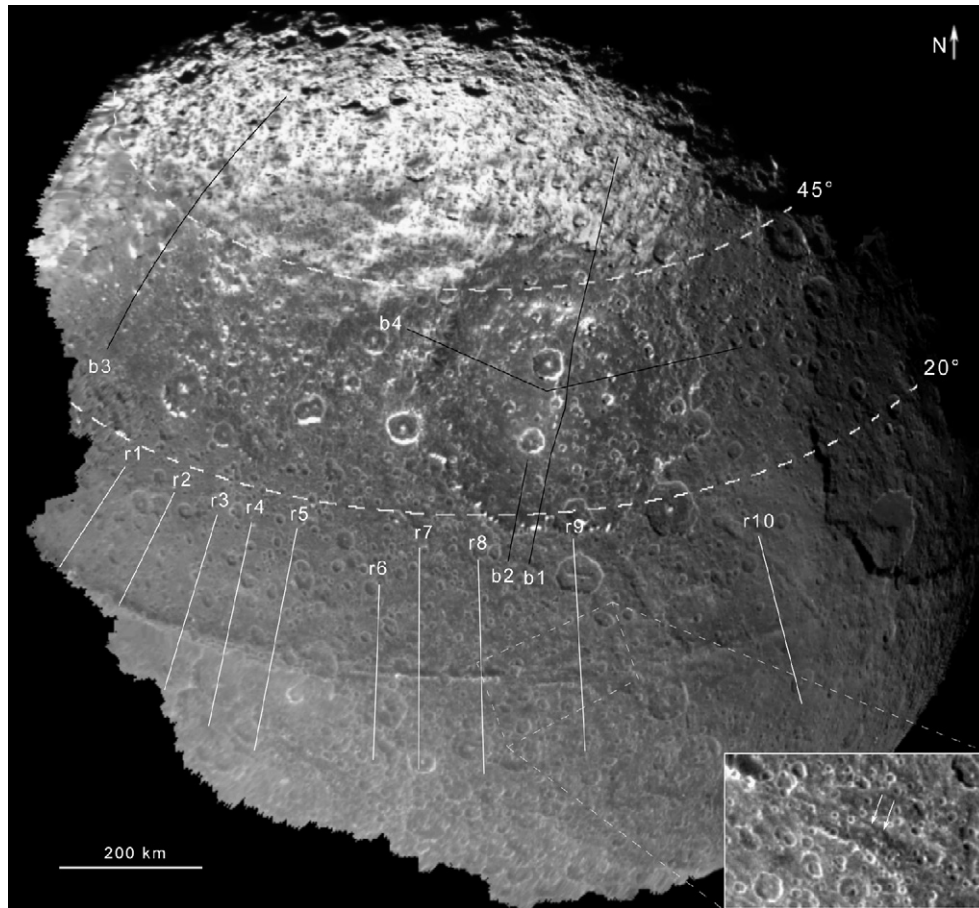


Fig. 6. Location of ridge profiles (r1–r10) and profiles (b1–b3) showing the relation between image brightness and topography (Fig. 7). Arrows in the inset mark a location where the inner ridge is in a depression but still intact.

The absence of the ridge in some places is clearly the result of impact erosion but in other places, e.g., the depression profiled by r9, it may be related to a more complex formational process than described above. Likewise, the change in the ridge shape from trapezoidal to triangular along the equator hints at a complex process. On the other hand, post-formational tectonic processes may have modified the shape of the ridge which could also explain the rather complex morphology.

A flat-topped ridge with sharp edges in places is hard to explain by Ip's theory, as is the fact that the ridge has upwarped the adjoining surface. Overall, the structure of the ridge is too complex to have been formed simply by deposition of putative ring particles.

4.3. Dark and bright terrains in relation to topography

Iapetus is well known for its albedo dichotomy between the leading side and the trailing side (Smith et al., 1981, 1982). The leading-side material is an order of magnitude darker than that on the trailing side and was suggested to be of exogenic origin (Vilas et al., 2004; Buratti et al., 2005). But high-resolution Cassini imaging has revealed that the leading side is not entirely dark. Here, we study the distribution of bright and dark terrains on the leading side in relation to topography. A detailed discus-

sion on the albedo dichotomy is given in a companion paper (Denk et al., in preparation).

4.3.1. Observations

Fig. 6 shows that northern latitudes $\geq 45^\circ$ are dominated by bright terrains and equatorial latitudes $\leq 20^\circ$ are predominantly dark. But bright areas are found at low ($\sim 20^\circ$) latitudes if slopes are north facing and steep (Fig. 7, compare topography and image brightness of the crater wall along b1: 29° slope with neighboring b2: 6° slope; b3: crater 1 & 2). Likewise, dark areas are found at high northern latitudes ($\sim 50^\circ$) if slopes are south facing (Fig. 7, b3: depression). East facing slopes, although as steep as the north facing slope, are dark (Fig. 7, compare b1 with b4). Dark material is preferentially collecting in topographic lows, e.g., at the bottom of craters (Fig. 8).

4.3.2. Implications

To darken the leading side of Iapetus solely from exogenic sources by an order of magnitude as compared to the trailing side would require unrealistically large amounts of dark material. Thus, a purely exogenic origin is unlikely (Denk et al., in preparation). Consistent with this hypothesis, our observations suggest that image brightness on Iapetus' leading side is controlled by sublimation with the dark material being a lag deposit originating from a bright icy substrate mixed with dark